

Soc Internship day 21

1. What is SOAR?

SOAR (Security Orchestration, Automation, and Response) is a cybersecurity platform that helps security teams automate repetitive tasks, integrate multiple security tools, and respond to incidents faster.

It combines:

- Orchestration – Connects different security tools (SIEM, EDR, Firewall, Email Security, etc.).
- Automation – Performs predefined actions automatically without manual intervention.
- Response – Executes incident response workflows to contain and remediate threats.

Example:

A phishing email is detected by the SIEM. SOAR can automatically:

1. Create an incident ticket.
2. Block the sender's email address.
3. Quarantine the email.
4. Notify the SOC analyst.
5. Collect related logs for investigation.

2. Simple Incident Playbook (Phishing Email)

Step	Action
1	SIEM detects a phishing email alert
2	SOAR validates the sender's reputation
3	Search for similar emails across the organization
4	Quarantine malicious emails automatically
5	Block sender's domain and IP address
6	Create an incident ticket
7	Notify the SOC analyst
8	Generate an investigation report
9	Close the incident after analyst verification

Steps

Phishing Alert

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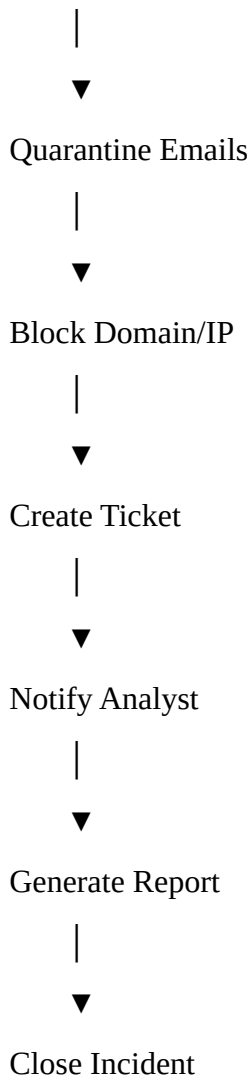


Validate Sender

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Find Similar Emails



3. Benefits of Automation in Cybersecurity

1. **Faster Response**
 - Incidents are handled within seconds instead of minutes or hours.
2. **Reduces Manual Work**
 - Analysts spend less time on repetitive tasks.
3. **Improves Accuracy**
 - Automated workflows reduce human errors.
4. **24/7 Operation**
 - Automation works continuously, even outside business hours.
5. **Consistent Incident Handling**
 - Every incident follows the same approved process.
6. **Increases SOC Efficiency**
 - Analysts can focus on complex investigations instead of routine tasks.
7. **Scalability**
 - Thousands of alerts can be processed automatically without increasing staff.

Example

Without automation:

- Analyst receives alert → investigates → blocks IP → creates ticket → notifies team (**10–20 minutes**).

With SOAR automation:

- Alert received → IP blocked → ticket created → analyst notified automatically (**less than 1 minute**).

This significantly reduces the **Mean Time to Respond (MTTR)** and strengthens an organization's overall security posture.